Date.....

Intermediate Value Theorem.

$f(x) = x^2 - 4x + 3$

(0, 2)

$f(0) = 3 > 0$

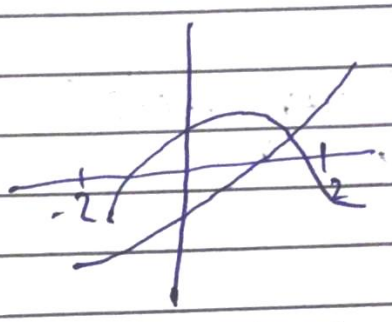
$f(2) = -1 < 0$

let $f(x) = x^3 + 3x + 1$ has exactly one real root between $[-2, 2]$

$f(-2) = -2^3 + 3x - 2 + 1 = -13 < 0$

$f(2) = 2^3 + 3x2 + 1 = 15 > 0$

$f'(x) > 0$



Homework:

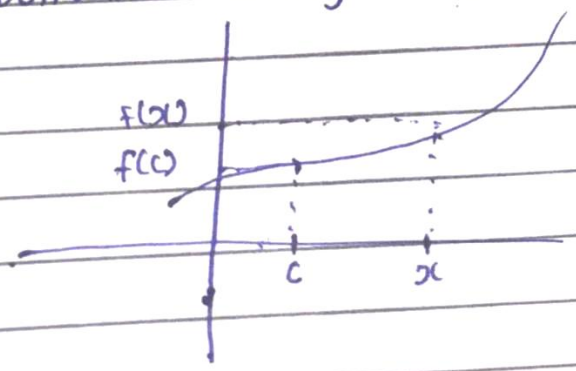
$f(x) = e^x - 3x$ has exactly one real root, (0, 1)

If $f(a) > 0$, $f(b) > 0$

$f(a) > f(b)$ $c \in [a, b]$ st. $f(c) > f(b)$

$f(a) >$

Derivative - Single Variable



x is any arbitary point

- Change = $f(c) - f(a)$
- Rate of change = $\frac{f(c) - f(a)}{c - x}$

Rate of change

$\frac{f(c) - f(x)}{c - x}$ = Difference quotient

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$\{x_n\}_{n=1}^{\infty} \to c$

$\left\{\frac{f(c)-f(x_n)}{c-x_n}\right\}_{n=1}^{\infty}$

For eg: $x_n = c - \frac{1}{n}$ - converge to c

$x_n = c - \frac{1}{n^2}$ - converge to c

$$f'(c) = \lim_{h \to 0} \left[\frac{f(c+h)-f(c)}{h}\right]$$

Q) $f(x) = x^2$

$f(c+h) = c^2 + h^2 + 2ch$

$$\frac{f(c+h)-f(c)}{h} = \frac{c^2+h^2+2ch-c^2}{h}$$

$$= \frac{h(h+2c)}{h}$$

$= h + 2c$

Q) $f(x) = x$

$$f'(c) = \lim_{h \to 0} \frac{f(c+h)-f(c)}{h}$$

$$= \frac{c+h-c}{h}$$

$$= \frac{h}{h}$$

$= 1$

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$f(x_n) \to c$ as $\{x_n\} \to c$ (Show) $\Rightarrow$ continuous at c.

$f(x)$ is diff. at c. $x_n \to c$

$\left\{\frac{f(x)-f(c)}{x-c}\right\} \Rightarrow f'(c) \Rightarrow (x_n - c) \to 0$

$\Rightarrow f(x_n) - f(c) \to 0$ - Hence f is continuous at c.

$f(x) = |x|$ - Show.

$x = 0$

$x_n = \frac{1}{n}$ - h_n

$x_n = \frac{-1}{n}$ - h_n

$f'(c) = \frac{|(c+h)| - c}{|h|}$

$= \frac{|h|}{h}$

If $h_1 = \frac{1}{n}$

$f'(c) = 1$

If $h_1 = \frac{-1}{n}$

$f'(1) = -1$

(Q) $f(x) = |x|$ at $x = 2$

(Q) $f(x) = |x|$ at $x = 1$. Show fn is differentiable or not.

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Q) $f(x) = \begin{cases} x & \text{if } x \geq 0 \\ x^2 & \text{if } x < 0 \end{cases}$ Is the function continuous? Is the fn differentiable?

$f(x) = \sin(x^2)$

$f'(x) = 2x \cos x^2$

Linear fn of f at $f(c) = g(c)$

let linear fn of $f(t) = \alpha + \beta t$

For linear approximation, 2 things should be satisfied

1. $f(c) = g(c)$
2. Rate of change of the error term $\left\{ \frac{|g(x) - f(x)|}{x - c} \right\} \to 0$

$$e(x) = g(x) - f(x)$$

$$\lim_{x \to c} \frac{e(x)}{x - c} \to 0$$

1. $g(c) = f(c) = \alpha + \beta c$

$\alpha = g(c) - \beta c$

$f(t) = g(c) - \beta c + \beta t$

$= g(c) + \beta(t - c)$

2. $\lim_{x_n \to c} \left\{ \frac{g(x) - f(x_n)}{x_n - c} \right\} \to 0$

$= \left\{ \frac{g(x) - g(c) + \beta(x_n - c)}{c - x_n} \right\} \to 0$

$= \left\{ \frac{g(x_n) - g(c)}{x_n - c} - \beta \right\} \to 0$

$= \frac{g(x_n) - f(c)}{x_n - c} - \beta = f'(x)$

derivative of fn g at c.

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$f(t) = g(c) + \beta(t-c)$ $\quad x_n \to 0$

$\Rightarrow f(c+h) = g(c) + g'(c)h \quad h_n \to 0$

$\Rightarrow g(c+h) - f(c+h) = e(h) \quad x_n = c+h_n$

$\Rightarrow g(c+h) \approx g(c) + f'(c)h$

2. $gof(x) = \sin(\tan(x^2))$

Do it ~~[illegible]~~ through the rule of chain rule.

Derivative Many variables

$f: R \to R \quad f'(c) = \beta$

f: f is differentiable at $x = c$

$\forall \{x_n\} \to c \quad \left\{\frac{f(x_n) - f(c)}{x_n - c}\right\} \to f'(c)$

$f: R^2 \to R \quad f(x_1, x_2 \ldots x_n)$

$\bar{x}_n = (\bar{x}_1, \bar{x}_2 \ldots \bar{x}_n)$

$x_1 - \bar{x}_1$

$x_2 - \bar{x}_2$

$$\frac{f(x_1, x_2) - f(\bar{x}_1, \bar{x}_2)}{\sqrt{(x_1 - \bar{x}_1)^2 + (x_2 - \bar{x}_2)^2}}$$

$x_1 - \bar{x}_1 = h_1$

$x_2 - \bar{x}_2 = h_2$

$h_1 \to 0$

$h_2 \to 0$

$$\frac{f(\bar{x}_1 + h_1, \bar{x}_2 + h_2) - f(\bar{x}_1, \bar{x}_2)}{\sqrt{h_1^2 + h_2^2}}$$

$$\frac{f(x_1 + x_1, \bar{x}_2) - f(\bar{x}_1, \bar{x}_2)}{h_1}$$

$h \to 0$

$$= \left.\frac{\partial f}{\partial x_1}\right|_{(\bar{x}_1, \bar{x}_2)}$$

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Date.....

$$\frac{\partial t}{\partial x_i}\Big|_{(\bar{x}_i)} = \lim_{t\to 0} \frac{f(\bar{x}_1, \bar{x}_2 \quad \bar{x}_i + t, \bar{x}_2) - f(\bar{x}_1 - \bar{x}_2)}{t}$$

$$= \lim_{t\to 0} \frac{f(\bar{x} + te_i) - f(\bar{x})}{t}$$

$z = f(x,y) \quad (\bar{x}_1, \bar{x}_2)$

$(\bar{x}_1\ \bar{x}_2, f(\bar{x}_1\ \bar{x}_2))$

~~$(z - \bar{z}) = \alpha$~~

$$g(x_1, x_2) - \bar{z} = \alpha_1(x_1 - \bar{x}_1) + \alpha_2(x_2 - \bar{x}_2)$$

$$\bar{z} = f(\bar{x}_1, \bar{x}_2)$$

$$g(x_1, x_2) = f(\bar{x}_1, \bar{x}_2) + (\alpha_1\ \alpha_2)\begin{pmatrix} x_1 - \bar{x}_1 \\ x_2 - \bar{x}_2 \end{pmatrix}$$

Property-2

$$\lim_{h\to 0} \frac{e(h)}{h} = 0$$

$$\lim_{\substack{x_1 - \bar{x}_1 \\ x_2 - \bar{x}_2}} \frac{f(x_1, x_2) - f(\bar{x}_1 - \bar{x}_2) - (\alpha_1\ \alpha_2)\begin{pmatrix} x_1 - \bar{x}_1 \\ x_2 - \bar{x}_2 \end{pmatrix}}{\|(x - \bar{x})\|} = 0$$

Let $x_1 - \bar{x}_1 = h_1$

$x_2 - \bar{x}_2 = h_2$

~~$\lim_{x_1 \to \bar{x}_1}$~~ $\lim_{\substack{h_1\to 0 \\ h_2 \to 0}} \frac{f(\bar{x}_1 + h_1, \bar{x}_2 + h_2) - f(\bar{x}_1, \bar{x}_2)}{\sqrt{h_1^2 + h_2^2}}$

$$= \frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot (\alpha_1 \cdot \alpha_2)\begin{pmatrix} h_1 \\ h_2 \end{pmatrix}$$

$$\lim_{h_1\to 0} \frac{f(\bar{x}_1 + h_1, \bar{x}_2) - f(\bar{x}_1, \bar{x}_2)}{h_1} = \alpha_1 = \frac{\partial t}{\partial x_1}\Big|_{(\bar{x}_1, \bar{x}_2)}$$

$$= \alpha_2 = \frac{\partial t}{\partial x_2}\Big|_{(\bar{x}_1, \bar{x}_2)}$$

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$(\alpha_1, \alpha_2) = \nabla f(\bar{x}_1, \bar{x}_2)$

$= \left(\frac{\partial f}{\partial x_1}\Big|_{(\bar{x}_1, x_2)} \cdot \frac{\partial f}{\partial x_2}\Big|_{\bar{x}_1, \bar{x}_2} \right)$

Example:-

$f(x,y) = |x|\,y$ at $(0,2)$

- Partial doesn't exist

~~$\frac{\partial f}{\partial x}\Big|_{(0,2)}$~~

To show fn is not differentiable show one counter example.

Homework:

$f: R^2 \to R$

$$f(x,y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

$x_1 \to \bar{x}_1$ $\quad x_1 = \bar{x}_1 + tg_1$

$x_2 \to \bar{x}_2$ $\quad x_2 = \bar{x}_2 + tg_2 \quad t \to 0$

$h_1 = tg_1$ $\quad g_1$ and g_2 are fixed.

$h_2 = tg_2$ $\quad t$ is fixed.

$t \to 0$

$\sqrt{x_1^2 + x_2^2} = 1$

$\lim_{t \to 0} \frac{f(\bar{x}_1 + tg_1, \bar{x}_2 + tg_2) - f(\bar{x}_1, \bar{x}_2)}{t}$ $\Rightarrow$ Derivative of fn f at the direction with norm 1

$= (\alpha_1 \alpha_2) \begin{pmatrix} g_1 \\ g_2 \end{pmatrix}$

$= \nabla(\bar{x}_1 \bar{x}_2) \cdot \begin{pmatrix} g_1 \\ g_2 \end{pmatrix}$

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$= \|\nabla f(\bar{x}_1, \bar{x}_2)\| \; \|(g_1, g_2)\| \cos\theta$

The direction and derivative of the function at $\bar{x}_1$ is direction of the vector.

$\|\nabla f(x_1, x_2)\| \; \|g_1, g_2\| \cos\theta$